

Computer Science

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Higher Level

Normalisation NF1, NF2 & NF3

Normalisation

Database **normalisation** is the process of organising the columns and tables of a relational database to minimise data redundancy.

Normalisation splits a large table into smaller tables and define relationships between them to increases the clarity in organising data.

Tables should have one purpose. Having the table serve many purposes introduces many of the challenges like: data duplication, data update issues, and increased effort to query data.

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Normalisation

If you have big database that is not normalised, people working with it most likely will face lots of problems. Performance will be bad, there are high chances of having multiple rows for one object, then, some day one row will change, and then you have a conflict.

Successfully normalized databases will increase performance and reduce storage.



Normalisation - 1st Normal Form - 1NF

Each form of Normalisation has a set of rule that defines it.

1st Normal Form Rules:

Each cell to be single valued

- Every cell stores on one piece of data, not multiple entries.



Name	Phone
Peter	555-1234
Paul	555-8765
Sara	555-1092



Name	Phone
Peter	555-1234 555-8456
Paul	555-8765
Sara	555-1092

Entries in a column are the same type

- Every column stores atomic values



Name	Eyes
Peter	Blue
Paul	Green
Sara	Hazel



Name	Eyes
Peter	Blue
Paul	Beautiful
Sara	Hazel

Rows uniquely identified

- Each row/record is unique and has a primary key



ID	Name
id1	Peter
id2	John
id3	Peter



Name
Peter
John
Peter

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Normalisation - 2nd Normal Form - 2NF

2nd Normal Form Rules:

The database must reach 1st Normal Form

- To proceed with 2NF, the database must have already gone through 1NF.

All attributes (non-key columns) are dependant on the key

Primary Key ✗ ✗ ✗ ✗

ID	Name	Phone	Eyes	Store	Store Phone	Item	Cost
id1	Peter	555-1234	Blue	H&M	666-9876	Suit	599
id2	Paul	555-8765	Brown	Massimo D	666-5467	Shirt	80
id3	Sara	555-1092	Hazel	Massimo D	666-5467	Trousers	120
id4	Peter	555-8458	Brown	H&M	666-9876	Tie	65
id5	John	555-3457	Hazel	H&M	666-9876	Suit	599

Primary Key ✓

ID	Name	Phone	Eyes
id1	Peter	555-1234	Blue
id2	Paul	555-8765	Brown
id3	Sara	555-1092	Hazel
id4	Peter	555-8458	Brown
id5	John	555-3457	Hazel

Primary Key ✓

Item	Store	Store Phone	Cost
Suit	H&M	666-9876	599
Shirt	Massimo D	666-5467	80
Trousers	Massimo D	666-5467	120
Tie	H&M	666-9876	65

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After NF2 there are two Primary Keys that come together as a compound key. With this compound key, you can still find all relevant information but with less redundancy.

✓ ✓

ID	Item
id1	Suit
id2	Shirt
id3	Trousers
id4	Tie
id5	Suit

Normalisation - 3rd Normal Form - 3NF

3rd Normal Form Rules:

The database must reach all requirements for 1st and 2nd Normal Form

- To proceed with 3NF, the database must have already gone through 1NF & 2NF.

All attributes (columns) can be determined only by the key in the table and no other column.

- 3NF involves the removal of transitive dependencies.

Primary Key ✗

ID	Name	Phone	Eyes	Store	Store Phone	Item	Cost
id1	Peter	555-1234	Blue	H&M	666-9876	Suit	599
id2	Paul	555-8765	Brown	Massimo D	666-5467	Shirt	80
id3	Sara	555-1092	Hazel	Massimo D	666-5467	Trousers	120
id4	Peter	555-8458	Brown	H&M	666-9876	Tie	65
id5	John	555-3457	Hazel	H&M	666-9876	Suit	599

Primary Key ✓

ID	Name	Phone	Eyes
id1	Peter	555-1234	Blue
id2	Paul	555-8765	Brown
id3	Sara	555-1092	Hazel
id4	Peter	555-8458	Brown
id5	John	555-3457	Hazel

Primary Key Foreign Key ✓

Item	Store	Cost
Suit	H&M	599
Shirt	Massimo D	80
Trousers	Massimo D	120
Tie	H&M	65

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In this case you do not want to enter the store phone number multiple times, so you create another table and eliminate redundancy. The new Primary key in this table becomes the Foreign Key in the previous.

✓

ID	Item
id1	Suit
id2	Shirt
id3	Trousers
id4	Tie
id5	Suit

Primary Key ✓

Store	Store Phone
H&M	666-9876
Massimo D	666-5467

Example - NF1, NF2 & NF3

I want to by a second hand car for driving to my new house in the country. Six people I know are selling cars, so I decided to make a database and do research before I purchase anything. I created a flat table database, but I think I need to normalise it.

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Mercedes Benz G-Wagon

Normalisation Example

Name	Phone	Brand	Type	Car	Manufacture's City	Cost	Year	Fuel	Colour
Adivije	+39 888 2121 +39 8882187	BMW	SUV	X3	Munich	4800	2006	Diesel	Silver
Tony	+39 777 0987	Audi	Sports	A1	Ingolstadt	Cheap	2010	Petrol	Black
Mike	+39 666 1693	BMW	Sedan	528i	Germany	6300	2010	Unleaded	Silver
Gavin	+39 555 9753	Mercedes	4 x 4	G Wagon	Stuttgart	9500	1981	Diesel	Scratched
Matthew	+39 444 3698	Toyota	Estate	Rav4	Toyota, Aichi	8600	approx. 2009	Petrol	Charcoal
Daniel	+39 333 0987	Mercedes	Truck	ML 350	Stuttgart	10800	2005	Petrol	Silver

Flat Database

✗

PK	Name	Phone	Brand	Type	Car	Manufacture's	Cost	Year	Fuel	Colour
Car1	Adivije	+39 888 2121	BMW	SUV	X3	Munich	4800	2006	Diesel	Silver
Car2	Tony	+39 777 0987	Audi	Sports	A1	Ingolstadt	6500	2010	Petrol	Black
Car3	Mike	+39 666 1693	BMW	Sedan	528i	Munich	6300	2010	Petrol	Silver
Car4	Gavin	+39 555 9753	Mercedes	4 x 4	G Wagon	Stuttgart	9500	1981	Diesel	Blue
Car5	Matthew	+39 444 3698	Toyota	Estate	Rav4	Toyota, Aichi	8600	2009	Petrol	Charcoal
Car6	Daniel	+39 333 0987	Mercedes	Truck	ML 350	Stuttgart	10800	2005	Petrol	Silver

✓ 1st Normal Form

PK	Name	Phone
Car1	Adivije	+39 888 2121
Car2	Tony	+39 777 0987
Car3	Mike	+39 666 1693
Car4	Gavin	+39 555 9753
Car5	Matthew	+39 444 3698
Car6	Daniel	+39 333 0987

Type	Brand	Car	Manufacture's	Cost	Year	Fuel	Colour
SUV	BMW	X3	Munich	4800	2006	Diesel	Silver
Sports	Audi	A1	Ingolstadt	6500	2010	Petrol	Black
Sedan	BMW	528i	Munich	6300	2010	Petrol	Silver
4 x 4	Mercedes	G Wagon	Stuttgart	9500	1981	Diesel	Blue
Estate	Toyota	Rav4	Toyota, Aichi	8600	2009	Petrol	Charcoal
Truck	Mercedes	ML 350	Stuttgart	10800	2005	Petrol	Silver

PK	Type
Car1	SUV
Car2	Sports
Car3	Sedan
Car4	4 x 4
Car5	Estate
Car6	Truck

2nd Normal Form

✓

PK	Name	Phone
Car1	Adivije	+39 888 2121
Car2	Tony	+39 777 0987
Car3	Mike	+39 666 1693
Car4	Gavin	+39 555 9753
Car5	Matthew	+39 444 3698
Car6	Daniel	+39 333 0987

Type	Brand	Car	Cost	Year	Fuel	Colour
SUV	BMW	X3	4800	2006	Diesel	Silver
Sports	Audi	A1	6500	2010	Petrol	Black
Sedan	BMW	528i	6300	2010	Petrol	Silver
4 x 4	Mercedes	G Wagon	9500	1981	Diesel	Blue
Estate	Toyota	Rav4	8600	2009	Petrol	Charcoal
Truck	Mercedes	ML 350	10800	2005	Petrol	Silver

PK	Type
Car1	SUV
Car2	Sports
Car3	Sedan
Car4	4 x 4
Car5	Estate
Car6	Truck

Brand	Manufacture's City
BMW	Munich
Audi	Ingolstadt
Mercedes	Stuttgart
Toyota	Toyota, Aichi

3rd Normal Form

✓

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Please note: The data redundancy for the Fuel and Colour would be addressed in 4NF.

Normalisation - Further Explanation

Visit the following YouTube clip for further explanation:

https://www.youtube.com/watch?v=UrYLYV7WSHM&t=827s

